

MASTER'S COMPREHENSIVE EXAMINATION

Part I

Theory (90 minutes)

December 9, 2006

Instructions: Answer all four questions. Open book and open notes.

1. Let Y_1, Y_2, Y_3 be a random sample of size 3 from a distribution with density

$$f_Y(y) = (\theta + 1)y^\theta, \quad 0 < y < 1, \quad \theta > -1.$$

- (a) Find a one-dimensional sufficient statistic.
- (b) Find the MLE of θ .

2. The P&A supermarket stocks three brands of yogurt; namely Yoplait, Bryers and LaYogurt. Bryers and LaYogurt are stocked equally but there are twice as many Yoplait. Assume each customer buys one yogurt, that purchases are made independently and after a sale the yogurt is restocked. Find

- (a) Probability that one of each yogurt is sold to the next three customers.
- (b) Probability that at least one of each yogurt is sold to the next four customers.

3. Let X_1, X_2, X_3 be three independent variables.

- (a) Suppose the variables are Poisson with parameters (1, 2, 4), respectively. Find the variance of

$$W = 2X_1 + 3X_2 - 5X_3.$$

- (b) Suppose next that the variables are normal with means (1, 2, 3) and variances (1, 2, 3), respectively. Find the probability that

$$W = 2X_1 + 3X_2 - 5X_3$$

exceeds -7 .

4. Let X_1, X_2 be a random sample of size 2 from a normal population with mean μ_x and variance σ_x^2 . Let Y_1, Y_2, Y_3 be a random sample of size 3 from a normal population with mean μ_y and variance σ_y^2 . Assume the samples are independent and $\sigma_x^2 = \sigma_y^2$.

- (a) Find a 95% confidence interval for $\mu_x - \mu_y$.
- (b) Find the MSE of the estimator

$$\left(\frac{X_1 + X_2}{2} - \frac{Y_1 + Y_2 + Y_3}{3} \right)$$

for $\mu_x - \mu_y$.